Universal gravitational energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the zero-at-infinity convention for gravitational potential energy.

SUCCESS CRITERIA - I can...

1. Use center-to-center distance.
2. Explain why negative potential energy does not mean negative kinetic energy.
3. Compute changes from final minus initial energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “potential energy” and “radius” to explain your model or result. Sentence starters are optional.

potential energy: energy associated with the configuration of interacting objects

radius: distance from the center to the edge of a circle or sphere

initial: at the beginning of the interval; subscript i

01 Warm-up: bring back an earlier idea

Net force rises linearly from zero to 10 N over 0.3 m. **Find** work and final speed for a 0.3 kg cart starting at rest.

Word help: work = energy transferred by a force along a displacement

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

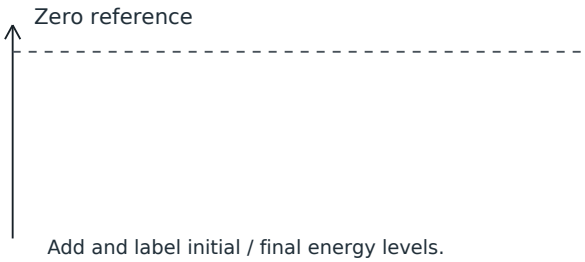
02 Worked example: explain the model

Reference: $U = -GMm/r$; $\Delta U = U_f - U_i$.
Outside a spherical source; choose U = 0 at infinity. Moving outward makes U less negative.

Worked reasoning: For a supplied source with GMm=120 J·m, potential energy is U=-120/r. At r=3 m, U=-40 J; at 6 m, U=-20 J. Moving outward raises U by 20 J.

Your turn: Draw an energy number line and mark the direction toward the zero at infinite separation.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label energies relative to U = 0 at infinite separation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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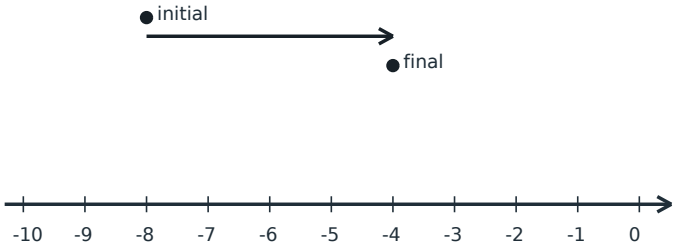
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

An object moves outward from r to 2 r. Does gravitational potential energy become more or less negative?
Draw a number-line comparison.

Sentence starter: The ... in my model represents ... . Its label shows ... .



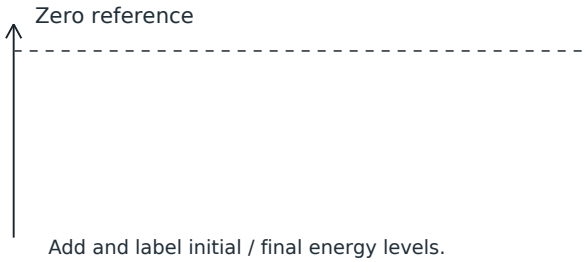
Example: potential energy (J)

Marzano check - circle: ① ② ③

04 Apply the physics

A 1000 kg satellite moves from r=7.0×10⁶ to 1.4×10⁷ m.
Use GM=4.0×10¹⁴ m³/s².
Find U at each point and ΔU.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student says “negative potential energy means the object has negative kinetic energy.” Correct the claim.

Sentence starter: I notice ... . This supports ... because ... .

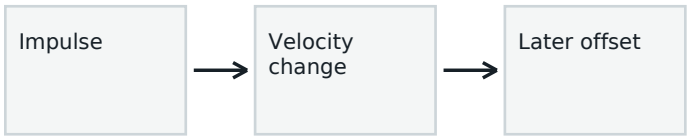
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A model target needs a 4 km sideways shift over 6×10⁶ s. Its mass is 8×10⁸ kg.
Estimate Δv and impulse; state the model used.
Label the quantities in the model and state its assumptions.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: ① ② ③

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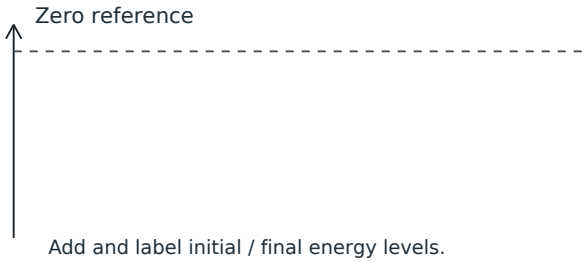
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

With GMm=60 J·m, move from r=5 to r=15 m.
Find work by gravity.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add and label initial / final energy levels.

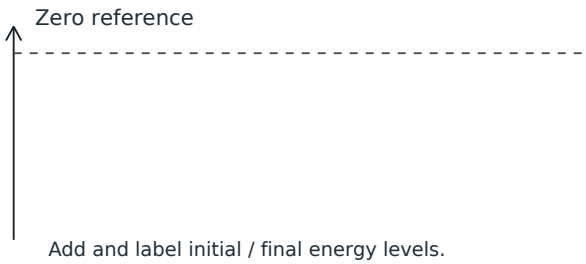
Label energies relative to U = 0 at infinite separation.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A mass moves from r=R to 3 R. Express ΔU and work by gravity in terms of GMm/R.
Compare with reaching infinity.

Sentence starter: Both ... . They differ in ... because ... .



Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: 1 2 3